

Amazon S3 and Amazon RDS Pricing Notes (us-east-1, sample reference data for this exercise)

Amazon S3 Storage Classes (approximate USD per GB-month):

- S3 Standard: \$0.023 per GB for the first 50 TB
- S3 Standard-IA (Infrequent Access): \$0.0125 per GB, plus a retrieval fee per GB accessed
- S3 Glacier: \$0.0036 per GB, intended for long-term archival with slower retrieval

S3 also bills separately for API requests (GET, PUT, LIST) and for data transferred out to the internet.

Amazon RDS Pricing:

- RDS pricing is driven primarily by the instance class (compute + memory) and the amount of allocated storage.
- Multi-AZ deployments roughly double the compute cost compared to a Single-AZ deployment of the same instance class, because a synchronous standby replica is also billed.
- Storage is billed per GB-month regardless of how much of the allocated storage is actually used, so significantly over-provisioning storage increases cost without a corresponding usage benefit.

General guidance for identifying anomalous spend:

- Compare the billed cost per unit (per hour for compute, per GB for storage) against the published reference rates above.
- A cost that is several times higher than the expected rate for the resource's size and type is a strong signal of anomalous or unexpected usage, and should be investigated before assuming it is simply high demand.